

# INFLATION

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# THE ROAD TOWARDS 2% INFLATION

12-month PCE inflation in May 2026:

- ▶ Headline: 4.1%
- ▶ Ex Energy Goods: 3.4%
- ▶ Core: 3.4%

Annualized PCE inflation through the year in May 2026:

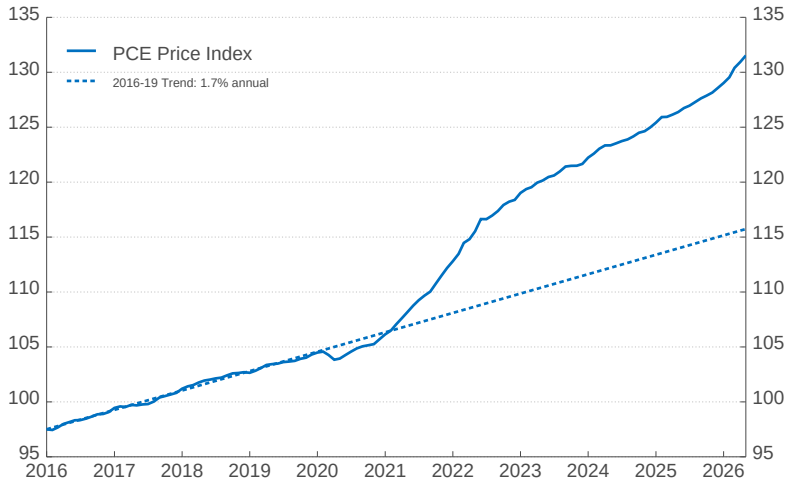
- ▶ Headline: 5.6%
- ▶ Ex Energy Goods: 4.1%
- ▶ Core: 4.2%

PCE Ex Energy Goods inflation trend for 2023-26: 2.8%

# Prices and Inflation

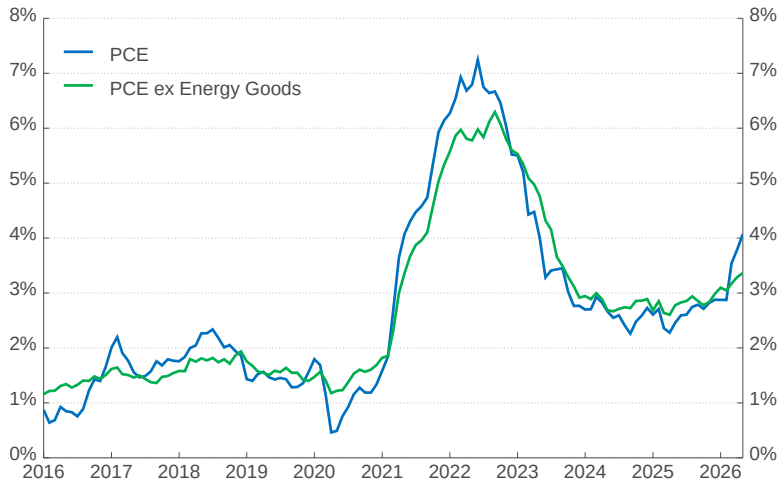
# PRICE LEVEL

PRICES DECLINE EARLY IN THE PANDEMIC AND RISE PERSISTENTLY ABOVE THEIR TREND AFTERWARDS



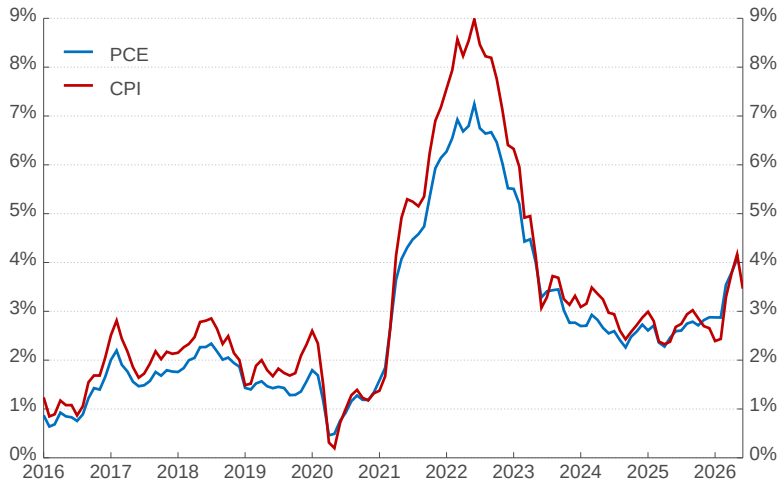
# INFLATION

INFLATION HAS REMAINED ABOVE 2% SINCE MARCH 2021—DYNAMICS AMPLIFIED BY ENERGY ▷ HISTORICAL SERIES



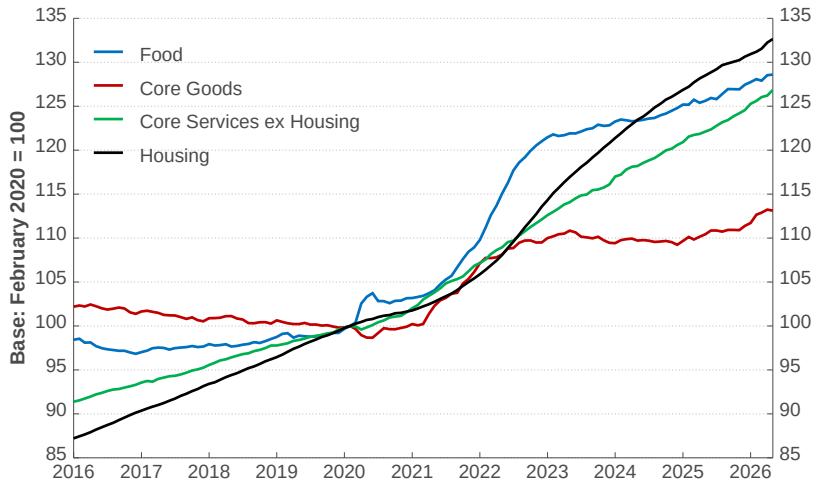
# COMPARING DYNAMICS IN PCE AND CPI

▷ EXCLUDING ENERGY



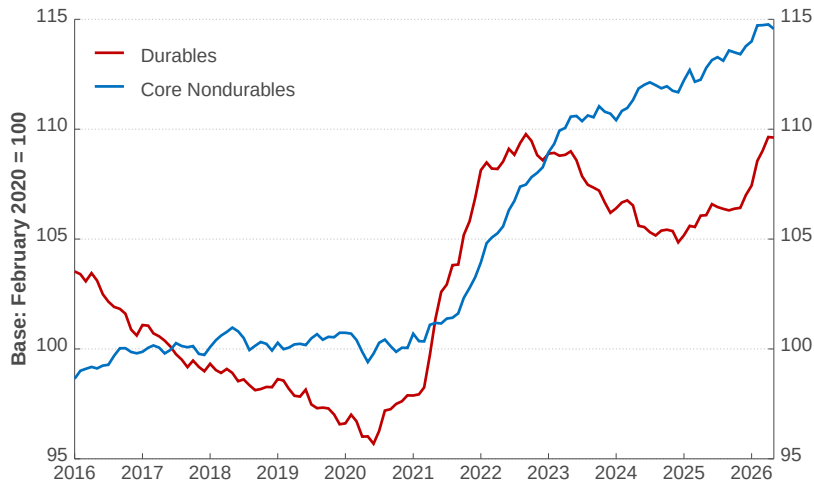
# COMPONENTS OF PCE PRICE INDEX

FOOD, CORE GOODS, CORE SERVICES EX HOUSING, AND HOUSING



# CORE GOODS: DURABLES VS. NONDURABLES

EFFECT OF TARIFFS EVIDENT ON PRICES OF CORE DURABLE GOODS—CORE NONDURABLES GROWING STEADILY





# INFLATION: AGGREGATE MEASURES

## ▷ CPI

### Annualized PCE Inflation Rates

|          | All Goods<br>& Services | Ex Energy<br>Goods | Excluding Food<br>& Energy (Core) |
|----------|-------------------------|--------------------|-----------------------------------|
| 2016–19  | 1.7%                    | 1.6%               | 1.7%                              |
| 2020     | 1.3%                    | 1.7%               | 1.5%                              |
| 2021     | 6.1%                    | 5.3%               | 5.2%                              |
| 2022     | 5.5%                    | 5.6%               | 5.0%                              |
| 2023     | 2.8%                    | 2.9%               | 3.1%                              |
| 2024     | 2.7%                    | 2.9%               | 3.0%                              |
| 2025     | 2.9%                    | 3.0%               | 3.0%                              |
| 2026 YTD | 5.6%                    | 4.1%               | 4.2%                              |

SOURCES: Bureau of Economic Analysis and author's calculations.

NOTE: The 2026 period is the year through May.

# INFLATION: MAIN COMPONENTS

## ▷ CPI

### Annualized PCE Inflation Rates by Categories

|          | Food  | Energy | Core Goods | Core Services<br>ex Housing | Housing |
|----------|-------|--------|------------|-----------------------------|---------|
| 2016–19  | 0.2%  | 4.2%   | −0.6%      | 2.2%                        | 3.4%    |
| 2020     | 4.0%  | −7.5%  | 0.1%       | 2.0%                        | 2.2%    |
| 2021     | 5.6%  | 30.3%  | 6.2%       | 5.2%                        | 3.7%    |
| 2022     | 11.1% | 6.7%   | 3.2%       | 4.9%                        | 7.8%    |
| 2023     | 1.5%  | −2.1%  | 0.0%       | 3.5%                        | 6.4%    |
| 2024     | 1.6%  | −1.0%  | −0.2%      | 3.9%                        | 4.7%    |
| 2025     | 2.1%  | 2.2%   | 2.0%       | 3.3%                        | 3.3%    |
| 2026 YTD | 2.2%  | 53.0%  | 3.8%       | 4.5%                        | 3.7%    |

SOURCES: Bureau of Economic Analysis and author's calculations.

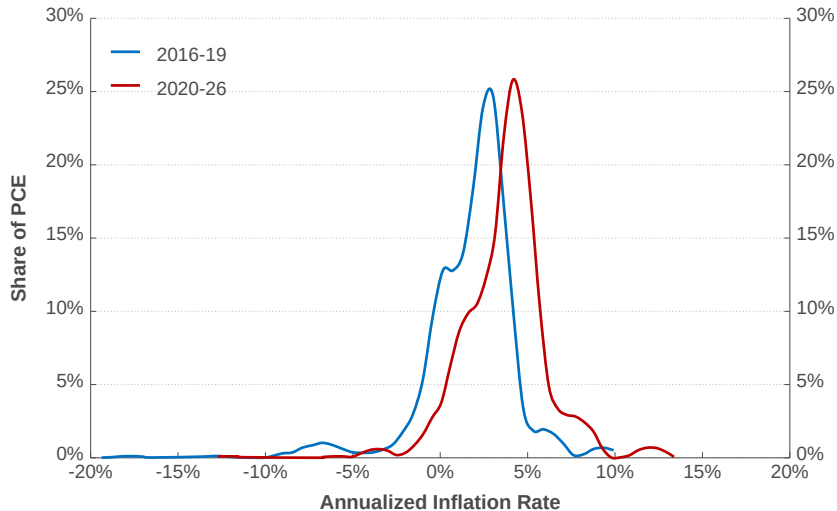
NOTE: The 2026 period is the year through May.

# Inflation Distributions

# INFLATION INCREASES HAVE BEEN WIDESPREAD

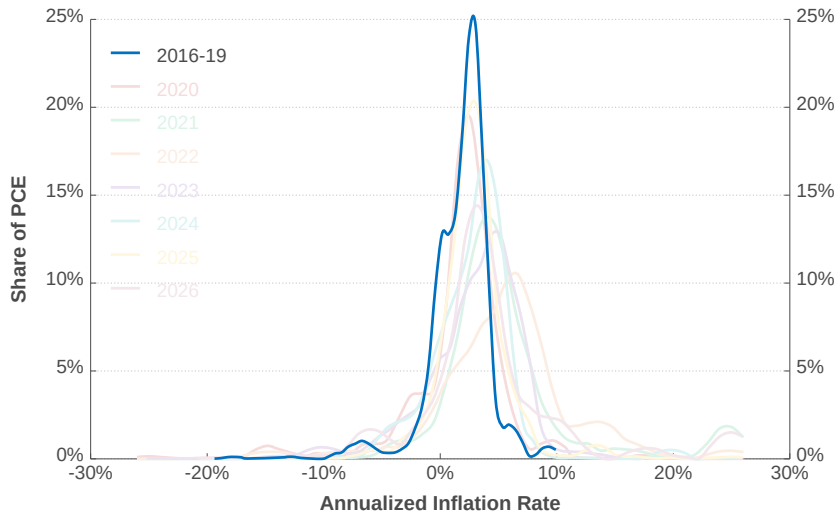
► METHODOLOGY

ESTIMATED DISTRIBUTION OF PCE INFLATION ► CORE



# INFLATION DISTRIBUTIONS: PROGRESSION FROM 2016-19 TO 2026

ESTIMATED DISTRIBUTION OF PCE INFLATION

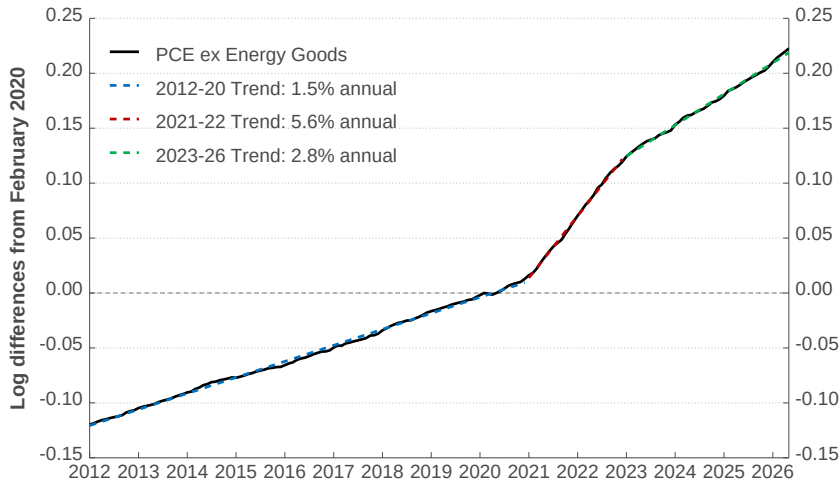


# Inflation Regimes

# PRICE LEVEL DYNAMICS: THREE REGIMES

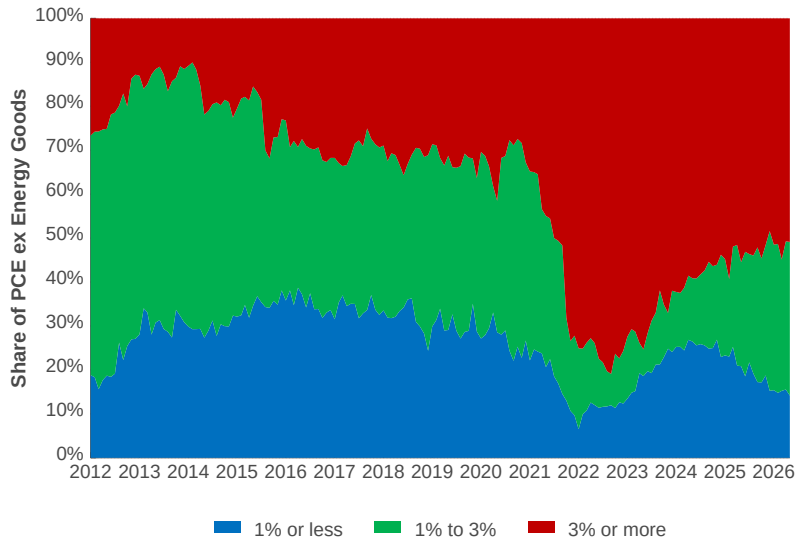
[► BLOG](#)

(1) BELOW-TARGET INFLATION: 2012-20 (2) GREAT INFLATION: 2021-22 (3) ABOVE-TARGET INFLATION: 2023-26



# EXPENDITURE SHARES BY INFLATION RATES

LARGE SHARE OF CONSUMPTION EXPENDITURES ON PRODUCTS WITH INFLATION RATES OF 3% OR MORE





# LONG-TERM INFLATION EXPECTATIONS

MARKETS VS. MODELS VS. FORECASTERS

## 10-Year Inflation Expectations — Period Averages

|                           | TIPS | Swaps | Cleveland | DKW  | SPF  |
|---------------------------|------|-------|-----------|------|------|
| 2012–20                   | 1.9% | 2.2%  | 1.7%      | 2.2% | 2.0% |
| 2021–22                   | 2.4% | 2.7%  | 1.9%      | 2.2% | 2.3% |
| 2023–26                   | 2.3% | 2.5%  | 2.3%      | 2.6% | 2.2% |
| Change 2012-20 to 2023-26 | 0.4% | 0.3%  | 0.5%      | 0.4% | 0.1% |

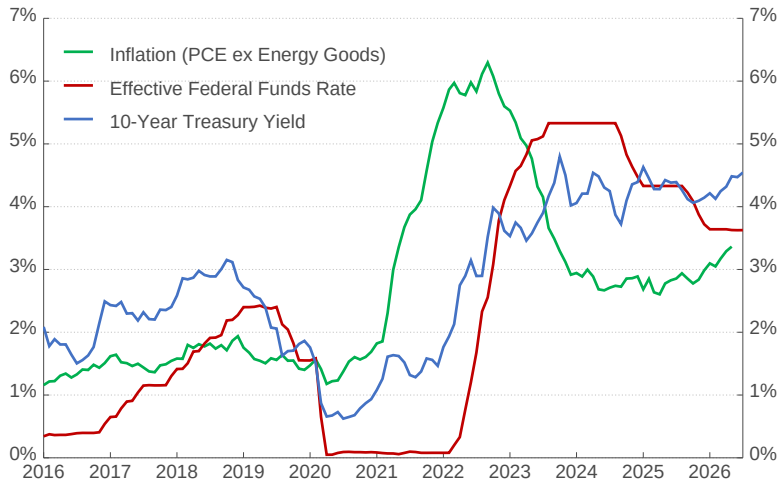
SOURCES: FRED, Federal Reserve Board, Federal Reserve Bank of Philadelphia, Bloomberg and author's calculations.

NOTE: The 2026 period is the year through May.

# Interest Rates

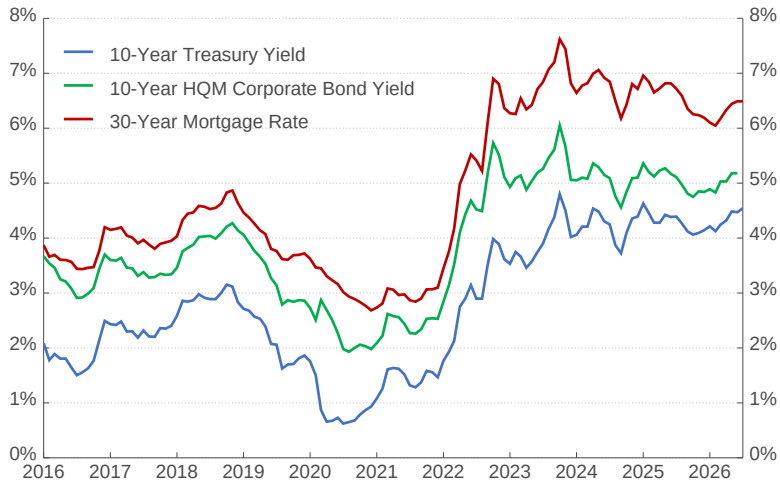
# MONETARY POLICY

EASING CYCLES IN SEPTEMBER-DECEMBER 2024 AND SEPTEMBER-DECEMBER 2025



# MONETARY POLICY AND LONG-TERM RATES

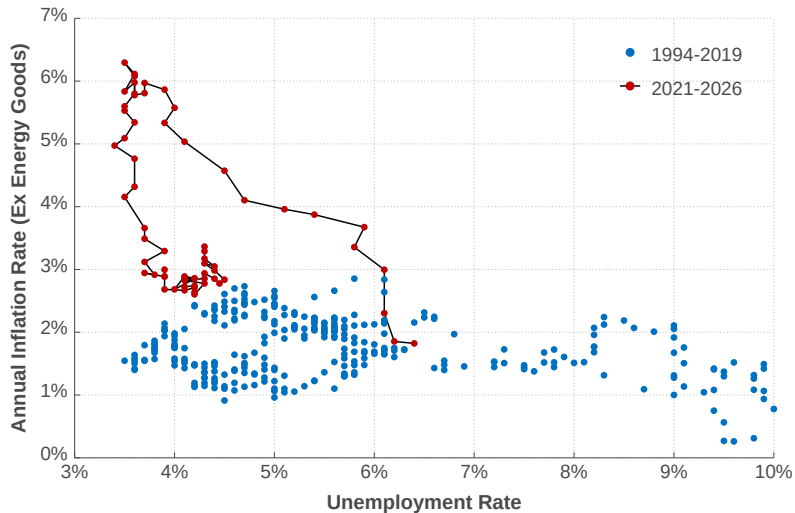
## ANNUAL RATES



# Phillips Curve

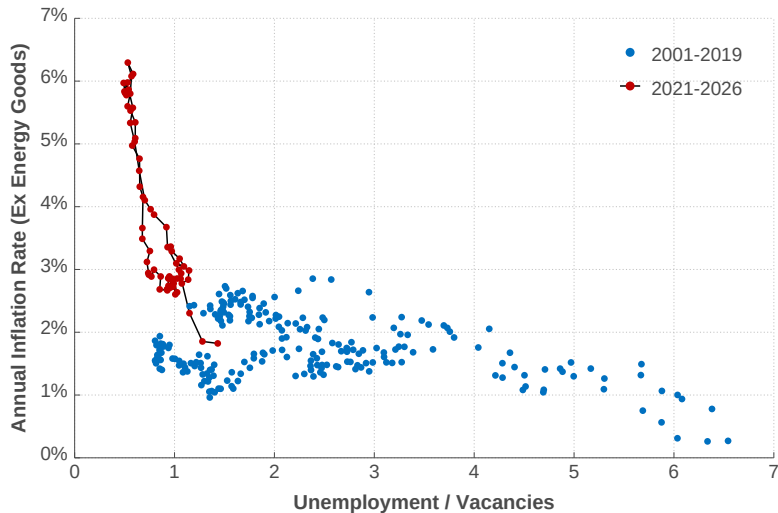
# THE PHILLIPS CURVE

INFLATION VS. UNEMPLOYMENT



# ALTERNATIVE PHILLIPS CURVE

INFLATION VS. UNEMPLOYMENT-TO-VACANCIES RATIO

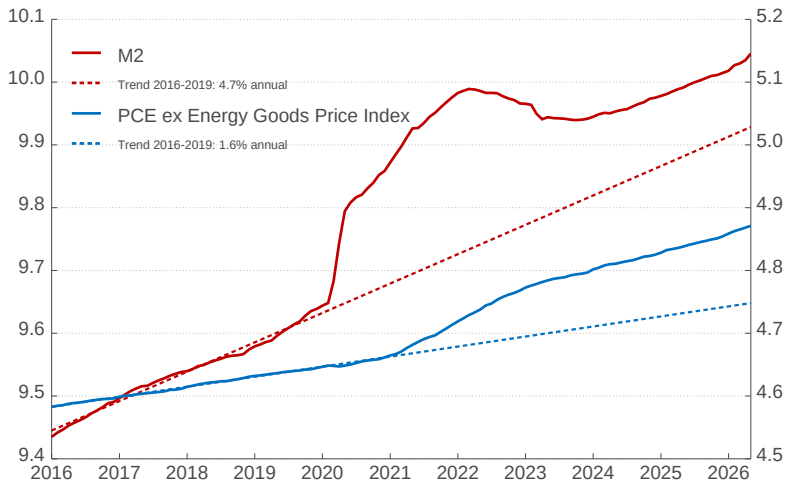


$$\mathbf{M} \times \mathbf{V} = \mathbf{P} \times \mathbf{Y}$$



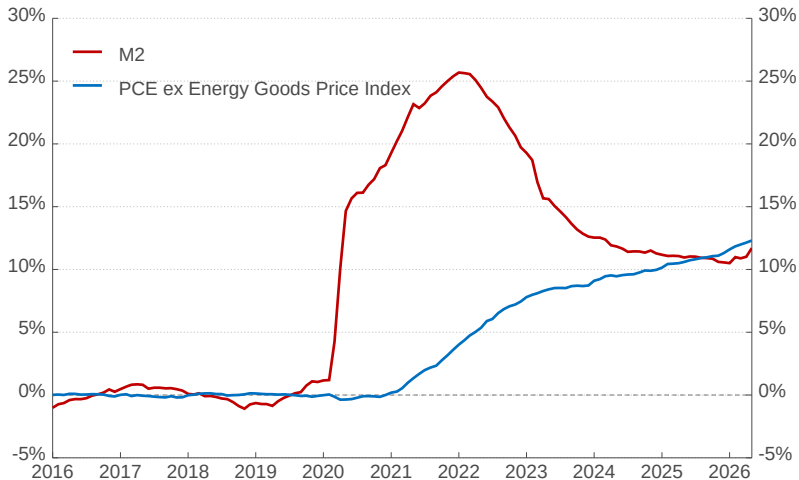
# MONEY VS PRICES: MONETARY EXPANSION DURING COVID

IN LOGS: M2 ON LEFT AXIS, PRICES ON RIGHT AXIS—M2 GREW AT SAME RATE AS DISPOSABLE INCOME PRE-COVID



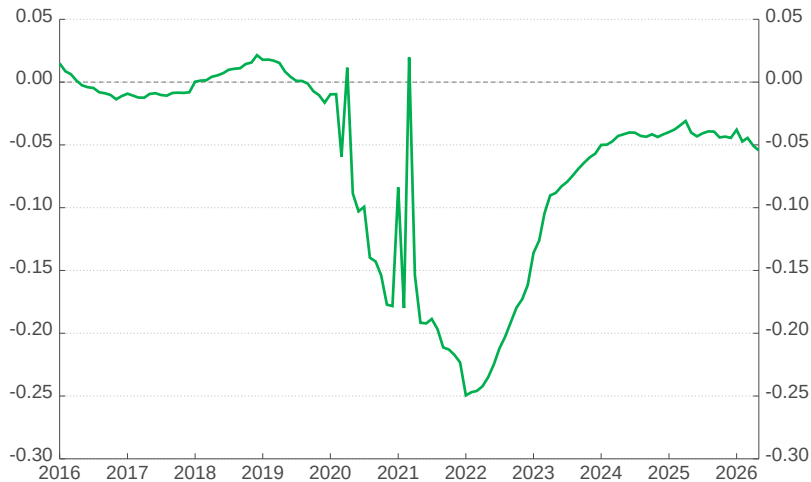
# PRICE INCREASE VS. MONETARY EXPANSION

DEVIATIONS FROM PRE-COVID TREND



# VELOCITY

$\text{LN VELOCITY} = \text{LN M2} - \text{LN DISPOSABLE PERSONAL INCOME}$  (BASE: 2016-19 = 0)



# Appendix

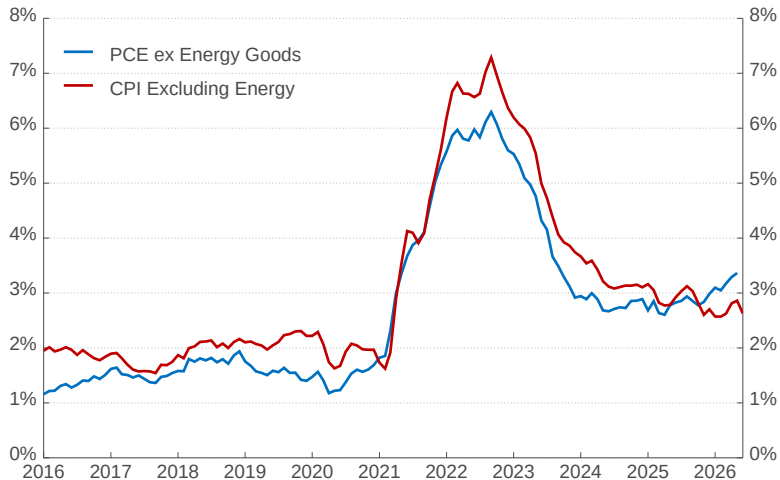
# CURRENT INFLATION IN HISTORICAL CONTEXT

PCE EX ENERGY GOODS ANNUAL INFLATION [▷ BACK](#)



# PCE EX ENERGY GOODS AND CPI EXCLUDING ENERGY

▷ [BACK](#)



# CPI INFLATION AGGREGATES

▷ [BACK](#)

## Annualized CPI Inflation Rates

|          | All Goods<br>& Services | Excluding<br>Energy | Excluding Food<br>& Energy (Core) |
|----------|-------------------------|---------------------|-----------------------------------|
| 2016–19  | 2.1%                    | 2.0%                | 2.1%                              |
| 2020     | 1.3%                    | 2.0%                | 1.6%                              |
| 2021     | 7.2%                    | 5.6%                | 5.5%                              |
| 2022     | 6.4%                    | 6.4%                | 5.7%                              |
| 2023     | 3.3%                    | 3.7%                | 3.9%                              |
| 2024     | 2.9%                    | 3.1%                | 3.2%                              |
| 2025     | 2.7%                    | 2.7%                | 2.6%                              |
| 2026 YTD | 4.1%                    | 2.6%                | 2.6%                              |

SOURCES: U.S. Bureau of Labor Statistics and author's calculations.

NOTE: The 2026 period is the year through June.

# CPI INFLATION COMPONENTS

▷ [BACK](#)

## Annualized CPI Inflation Rates by Categories

|          | Food  | Energy | Core<br>Commodities | Services<br>Less Rent | Rent of<br>Shelter |
|----------|-------|--------|---------------------|-----------------------|--------------------|
| 2016–19  | 1.2%  | 3.9%   | −0.2%               | 2.2%                  | 3.3%               |
| 2020     | 3.9%  | −7.0%  | 1.6%                | 1.4%                  | 1.9%               |
| 2021     | 6.0%  | 30.3%  | 10.7%               | 3.9%                  | 4.2%               |
| 2022     | 10.1% | 6.9%   | 2.1%                | 7.5%                  | 7.6%               |
| 2023     | 2.7%  | −1.9%  | 0.1%                | 3.5%                  | 6.2%               |
| 2024     | 2.4%  | −0.3%  | −0.7%               | 4.0%                  | 4.6%               |
| 2025     | 3.0%  | 2.1%   | 1.4%                | 3.4%                  | 3.1%               |
| 2026 YTD | 2.8%  | 25.0%  | 0.1%                | 3.5%                  | 3.6%               |

SOURCES: U.S. Bureau of Labor Statistics and author's calculations.

NOTE: The 2026 period is the year through June.



# CORE INFLATION INCREASES HAVE BEEN WIDESPREAD

ESTIMATED DISTRIBUTION OF CORE PCE INFLATION [▷ BACK](#)

